

Mathematical Intuition:-

Dataset

Point	Sweetness	Crunchiness	Class
A	2	1	Apple
B	3	2	Apple
C	6	5	Mango
D	7	6	Mango
E	5	4	Mango
?	4	3	???

Euclidean Distance:-

$$d(x, y) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

To classify new data point, we need to know the nearest distance datapoints by majority vote, we assign a new data point a class.

Calculations:-

Calculate Distance from every point.

$$A = \sqrt{(4-2)^2 + (3-1)^2} = \sqrt{4+4} = \sqrt{8} = 2.828$$

$$B = \sqrt{(4-3)^2 + (3-2)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142$$

$$C = \sqrt{(4-6)^2 + (3-5)^2} = \sqrt{4+4} = \sqrt{8} = 2.828$$

$$D = \sqrt{(4-7)^2 + (3-6)^2} = \sqrt{9+9} = \sqrt{18} = 4.2426$$

$$E = \sqrt{(4-5)^2 + (3-4)^2} = \sqrt{1+1} = \sqrt{2} = 1.4142$$

Decide K,

We take K for voting, use it randomly.

$$K = 3$$

Sort 3 smallest distances:-

We take only 3 shortest distance data points and sort in ascending order.

$$B = 1.4142 \quad (\text{Apple})$$

$$E = 1.4142 \quad (\text{Mango})$$

$$A = 2.828 \quad (\text{Apple}) \quad (\text{its our choose to take A \& C both are same.})$$

Majority Vote:-

$$\text{Apple vote} = 2$$

$$\text{Mango vote} = 1$$

So, the new data point goes into the "Apple class" because of its majority vote.

Day: _____

Date: _____

New Data Set

Point	Sweetness	Cronchiness	Class
A	2	1	Apple
B	3	2	Apple
C	6	5	Mango
D	7	6	Mango
E	5	4	Mango
F	4	3	Apple